Pioneer academics

The Practical Challenge of Writing Secure Code Syllabus

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Research Concentration Description:

Despite the rapid advances in technology, cyberattacks continue to rise. In addition to the shortage of cybersecurity experts, another challenge is to have developers write code that is secure to begin with. When investigating attacks, it is often the case that a bug exists in the code due to insecure coding/programming practices. Research shows that considering cybersecurity requirements at the early stages of development helps prevent vulnerabilities and reduce errors by 90%. Unfortunately, secure coding practices are often ignored by programmers who rely on the testing phase to find bugs and vulnerabilities.

In this research concentration, we will first learn about cybersecurity basics and important secure coding practices that are essential for writing secure programs. In the second portion of the program, each scholar will develop independent research projects that include an examination of security best practices. Understanding and analyzing code is required for this concentration, and therefore, basic familiarity with programming concepts and a programming language (C/C++, Java, Python) is required. Scholars are expected to follow a rigorous research method as guided by ACM's SIGSOFT Empirical Standards for Software Engineering. Several papers and online book chapters will be assigned for readings, and scholars will be given guidance to learn cybersecurity through online CTF gaming if interested in sharpening their cybersecurity skills.

Academic Integrity:

The Pioneer Research Program provides every scholar the opportunity to develop their thinking, writing, and research skills. Scholars are expected to make the most of this opportunity by practicing those skills in all of their work, including the final paper.

It is likely, however, that you might wish to use the assistance of ChatGPT or AI tools in your work during this program. While ChatGPT/AI is not prohibited, I do expect you to use these tools with discretion and moderation. Your final paper is to be a representation of your work and integrity. **Appropriately citing any text generated by ChatGPT/AI, even when you change the wording, is required and represents responsible scholarship**. Citation guidelines can be found here:

- APA Style
- MLA Style
- Chicago Style

Here is a guideline for the use of AI/ChatGPT in this class for our assignments and final paper:

Permitted uses of AI	Idea development; Source suggestions
Uses that require permission	Organizing themes; Constructing an outline; All other uses
Not permitted uses of AI	Dataset explanation; Drafting summaries; Comparative analysis; Drafting sections

Additional Guidelines

When use of AI is permitted, **you must acknowledge such use** by:

- identifying the specific tool used and the date on which it was used;
- providing a copy of the prompt(s) used to generate output;
- "...describ[ing] how you used the tool in your methodology section or in comparable sections of your paper. For literature reviews, you [should] describe how you used the tool. In your text, provide the prompt you used and then any portion of the relevant text that was generated in response" [2]
- formally citing (in both in-text and reference citations) any work generated by AI, including placing verbatim (word-for-word) AI output in quotation marks and citing such output when you paraphrase it (put it in your own words), just as you would when using any other reference material, in written assignments.

When use of AI is permitted, you should be aware of the limitations of these tools, including, but not limited to the following:

- **The quality of the prompts (inputs) you provide directly influences the quality of the results (output) the tool provides.** It takes practice, and often multiple, iterative attempts, to refine prompts that yield better outcomes.
- Regardless of the quality of your prompt, **you should not trust the output of AI tools**. Assume that facts and figures provided are wrong unless you either know them to be true or can verify their accuracy by consulting one or more other reliable sources. (For example: Lawyers submitted a motion generated by ChatGPT in a recent court case in New York, and the AI tool cited "bogus" cases in its output.[3])
- **You are responsible for the accuracy of any work you submit**, and thus, you will be held responsible for any errors or omissions generated by the tool that you do not catch and correct. (The lawyers in the case referenced above face sanctions for not checking the accuracy of the motion they filed in court.)
- You should be thoughtful about your use of such tools, even when it is permitted. Think carefully about the best ways to use the tools to accomplish your learning objectives. [4]

I and Pioneer Academics reserve the right to use AI-detection tools when evaluating assignments submitted in fulfillment of the Research Program requirements. **Any unauthorized use of ChatGPT or similar AI tools, without appropriate citation throughout your paper, constitutes plagiarism and will be treated as such.** Plagiarism of this type can result in a grade penalty, a failing grade, and/or removal from the Pioneer Research Program.

IMPORTANT: Please read the Academic Integrity Policy (Student Enrollment Agreement)

[1] Thanks to Dr. Jody L. Kerchner for this language, which is used with her permission.
[2] McAdoo, T. (2023, April 7). How to cite GPT. American Psychological Association. Retrieved July 22, 2023, from https://apastyle.apa.org/blog/how-to-cite-chatgpt.
[3]
https://www.cnbc.com/2023/05/30/chatgpt-cited-bogus-cases-for-a-new-york-federal-court-filing.html
[4] This policy is modeled on one developed by Dr. Ethan Mollick and Dr. Lilach Mollick and is used with their permission.

Pioneer LMS Link: pioneeracademics.schoology.com

Required Program Material:
Readings will be available online and linked on Slides and Schoology.

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Learning Goals:

Students in this program should be able to:

- Demonstrate understanding of the scientific method used to explore a problem, including observation, hypothesis development, measurement and collection of data, experimentation, and evaluation of evidence.
- Collect and analyze data as appropriate in CS and cybersecurity.
- Utilize current scientific literature to formulate new research questions.
- Construct a reference list and appropriately cite sources.
- Communicate effectively orally and in writing throughout the duration of the program, and give a final oral presentation about the research.

Grading Policy:

Grading Scale:

A	A-	B+	B	B-	C+	C	C-	D	F
93+	92-90	89-87	86-83	82-80	79-77	75-73	72-70	69-60	<60

Grade breakdown:

80%: Final Paper

The grade for the final paper is broken down as follows:

- *10%: Proposal (Purpose, Questions you will explore, outlined methodology)*
- *10%: Proper use and formatting of bibliography and citations*
- *5%: first draft with an introduction and background that includes the literature review*
- *5%: second draft with methodology and results*
- *5%: One peer feedback evaluation*
- *45% Final paper*

10%: Final Presentation

The final session will include final presentations. Each student is assigned a 15-minute slot during the final session. Students are expected to present in the first 10 minutes and use the remaining 5 minutes to answer questions.

→ 5% is for presentation slides (e.g., using PowerPoint or Google drive slides).

- Make sure that your slides are well formatted and readable. Include slide numbers and avoid typos/grammar/language mistakes.

→ 5% is for delivery during the session. Please use the following guidance:

- Focus on brevity and distill your topic into the most important elements. You need to make sure that you deliver the main takeaway and findings from your project to your peers.
- A practice run before class to test out the length of your presentation is a good strategy.
- Please remember to cite your sources in your slides. You may cite either at the bottom of each slide or using a dedicated source slide.

10%: Written response to discussion questions
Students need to show active engagement with the material presented in group sessions. This course heavily relies on reading papers and conducting research. Therefore, students are expected to show up to class prepared to have meaningful discussions and respond to the discussion questions of the session. Students are expected to come prepared to address the discussion questions presented in class by submitting a short written response of 1-2 paragraphs. Students are expected to share their ideas using their own wording/phrasing and should cite resources that helped them for the response to the question. Discussion questions are required for group sessions 1, 2, and 4.

Assignments and Milestones:

In group sessions, we will go over topics that would help you navigate your research paper. However, there will be some suggested resources to read, but for the most part, you will not have an officially assigned reading. It is part of this learning experience to learn how to search for information and read multiple resources, so you are asked to find your own resources to answer the discussion questions in class.

Session 1 (Group Meeting): Introduction to computer security and secure programming. In this session, we will go over the history of computer security and the evolution of the programming language while highlighting issues that make secure coding a challenging problem in today's code. This session will also review paper reading and writing strategies that are helpful for you to start your research.

> ***Discussion questions*** *in Session 1:* What does computer security mean to you? What do secure coding practices mean to you?

Session 2 (Group Meeting): Input validation and its role in avoiding code injection. In this session, we will explain common code injection vulnerabilities in SQL, XSS, and in Buffer overflows. We will explain the importance of input sanitization and how checking user input is an essential step that often gets missed in writing secure code.

> ***Discussion question*** *in Session 2:* Why do you think existing code does not follow secure coding practices for input validation, and we continue to see vulnerabilities?

Session 3 (Group Meeting): C language vulnerabilities. In this session, we will explain in depth some vulnerabilities specific to the C language. We will revise buffer overflow from the previous session and provide examples of integer issues and format strings. We will talk about some defense strategies, secure practices, and adoption barriers.

Session 4 (Group Meeting): Passwords, authentication, and data encryption in software. In this session, we will go over some cryptographic basics to explain how data encryption is important to protect sensitive data. We will explain the difference between encryption and hashing. Then, we will provide examples of how access control can be circumvented with weak programming practices that can weaken authentication (e.g., hard-coded passwords, unencrypted sensitive information, etc.). Finally, we will discuss ideas for students to structure their research proposal.

> ***Discussion question*** *in Session 4:* What makes passwords critical? Can you think of ways attackers can use to obtain user passwords?

Session 5 (Individual Meeting): Students are expected to come with a draft/outline of their project proposal to discuss with the instructor.

Session 5 (Individual Meeting): Students are expected to come prepared with six project ideas, each described in a short paragraph. During the session, students will discuss these ideas with the instructor to select a project. By the end of the session, students will write their project proposal and review it with the instructor.

Assignment to be completed 24 hours after this session:
Write a revised project proposal that lists your research question(s) and rationale/purpose for the research project. You also need to give an overview of your methodology, i.e., how you are going to approach the research question. Your revised proposal should incorporate the feedback you received from the instructor in the individual session. Provide bibliographic citations.

Session 6 (individual Meeting): Pre-session preparation: Students must complete a thorough literature review.
Session activities:

- Review the literature review and background with the instructor
- Discuss how the literature informed research approach
- Create a paper outline with instructor feedback

Assignment to be completed 24 hours after this session:
Refined fine-grained paper outline.

Session 7 (individual Meeting): Discuss paper in progress. Students are expected to come with a draft of the methodology and highlights of their preliminary results. Review the methodology and preliminary results with the instructor during the session.

Assignment to be completed for the session:
Write a revised methodology section of your paper. Your revised version should incorporate the feedback you received from the instructor in the individual session. Provide bibliographic citations.

Session 8 (Individual Meeting): Discuss paper in progress. Students are expected to come with a draft of the results and discussion sections. Review the results and discussion sections.

Assignment for this session:
Write a revised results section of your paper and start writing the discussion section. Your revised version should incorporate the feedback you received from the instructor in the individual session. Provide bibliographic citations.

Session 9 (Individual Meeting): Final paper discussion and review. Students are expected to come with a draft of the final paper and discuss their plan for the final presentation.

Assignment for this session:

- Prepare your presentation
- Continue finalizing the paper that will be due two weeks after Session 10.

Session 10 (Group Meeting): Final Presentations. In this session, students are expected to present in class and answer discussion questions about their research. Students can use feedback and comments from the group to finalize their final paper. Instructions for the final presentation are listed above under the grading section.

Final Paper Requirements:

The Final paper is 45% of the grade. Students are expected to submit a 15-page single-column paper using the IEEE style for citation. Students can use tools like Zotero and Mendeley that have MS Word and Google Docs plug-ins, and that would make the citation task easier.

IMPORTANT: Please read the Final Paper Topic Approval Policy

Final Paper Deadline: Ten days after the final session.

Peer Feedback:

The professor will assign each scholar a peer partner. Scholars will exchange their papers during week 8. Students will also arrange for a Zoom meeting during which they provide their peers with feedback, based on the comments made on the Peer Feedback Evaluation form. Please contact our Program Cohort Advisor if you need assistance in setting up a time to meet with your peer partner via Zoom. This meeting will be in addition to the 10, regularly scheduled individual and group Pioneer sessions.

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Pioneer Syllabus Policies

Contact Method:

All communication outside of meeting times will be conducted on the Pioneer Learning Management System (LMS). As per the student agreement, use of email to communicate with the professor mentor during the program is strictly prohibited. If you have problems using the Pioneer LMS, contact your Pioneer Cohort Advisor.

Pioneer Schoology Link: pioneeracademics.schoology.com

Pioneer Writing Center:

The Pioneer Writing Center is a fully online, asynchronous resource dedicated to supporting scholars in the Pioneer Research Institute on their research papers. The Writing Center serves as an essential resource to help scholars in the following areas:

- Grammar and language usage
- Structure and logic
- Basic style and presentation

All scholars may use the Pioneer Writing Center to review up to four drafts of their paper. Scholars are REQUIRED to submit at least one draft to the writing center. The Pioneer Writing Center generally provides feedback within two days of submission. Scholars can submit drafts by sending their work to writingcenter@pioneeracademics.com or completing the Google form on Schoology.

Turnitin:

The Turnitin tool is a part of our dedication to academic integrity. The software will compare the submitted paper against websites, books, journal articles, other student papers, etc., and highlight any sections of the paper that could potentially match such materials. Scholars must check their papers when submitting drafts to the writing center and before submitting the final version for grading by using the "Check your paper for plagiarism here" assignment, also located in the Pioneer Scholar Central section on Schoology. All final papers must be submitted through the "Submit your final paper here" Turnitin assignment in the Course Materials section of the LMS.

Pioneer Support Curriculum:

The Pioneer Support Curriculum is a series of seminars, workshops, and modules designed to support you during critical junctures of your Pioneer journey and to ensure that you are prepared for the rigors of undergraduate-level research. The curriculum includes your Cohort Countdown, the Starting Strong series, Research Seminars, and Alumni Workshops. Each portion includes live sessions and corresponding asynchronous resources. Scholars are required to attend a live session of each event and to review all asynchronous materials. Registration information for live sessions will be posted by the PCA. All asynchronous materials are available in the Pioneer Cohort Central folder on Schoology. It is *your* responsibility to review the folder carefully and regularly, as information may change or be added as you transition through the different phases of your experience.

Technical Support:

For any technical issues relating to program software, the Pioneer LMS or the Oberlin College library system that are not urgent, scholars should contact their Pioneer Cohort Advisor. For all issues that require immediate attention call the Pioneer Support Line at +1 (855) 572-8863. A Pioneer staff member will respond to help resolve any problem quickly.

Rescheduling Sessions:

Each Pioneer Scholar has the privilege of requesting to reschedule one individual session with a valid reason. No other reschedule will be allowed except in an emergency, such as a sudden illness or family emergency. Each reschedule request will be evaluated on a case-by-case basis. Any rescheduling request must be submitted to Pioneer using the “Reschedule Request Form,” available in Course Materials on Schoology, at least 48 hours before the relevant session. Scholars are responsible for being aware of their schedule and any applicable time zone conversion and ensuring they have a strong enough internet connection before the session. If the missed session was due to a sudden emergency, Pioneer will consider rescheduling on a case-by-case basis, but the rescheduling request cannot be guaranteed.